

FOUNDATIONS AND ASSEMBLY FOR THE IUT- abc FORMALIZATION CONCRETE LEGENDRE INPUT, TEN-COMPONENT INITIAL-THETA DATA, MULTIRADIAL SOURCE INGREDIENTS, AND THE REFINED NON-CIRCULAR BOUNDARY

CHATGPT

ABSTRACT. This note records the next construction layer in a Lean formalization programme for the IUT route to the abc conjecture. For a positive coprime triple $a + b = c$, it constructs the rational Legendre parameter $\lambda = a/c$, the associated Weierstrass equation, and proves the discriminant is nonzero. It then separates the ten conceptual components of public initial theta-data into dependent Lean types and gives an exact assembly map into the public `Iut.InitialThetaData` record. Existing public constructors for Tate parameters and torsion fields are incorporated.

The IUT III stage is decomposed into an explicit multiradial-output type, a distinguished native branch, public packet realizations, the exact native volume identity, and monotonicity. These data assemble into the already formalized public `GeneratedNativeSource`. The IUT IV stage is likewise decomposed into the uniform Theorem 1.10 estimate, the q -height comparison, and the actual different and conductor comparisons. These six ingredients assemble into the non-circular downstream bridge and imply the logarithmic abc conjecture.

The note does not claim an unconditional proof. The remaining work is inhabitation: the large-image/admissible-prime family, the actual orbicurve/core/tempered family, the Hodge-theater/Frobenioid multiradial realization, and the uniform IUT IV geometric estimates have not been constructed as closed Lean terms.

Formal status

The accompanying package contains 47 Lean source files and no intentional `sorry`, `admit`, or user-declared `axiom`. It was assembled offline and has not been kernel-built in the present runtime. It proves

$$\text{RefinedFourStageProgram} \implies \text{ABCConjecture},$$

but does not prove `Nonempty RefinedFourStageProgram`.

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1. THE RATIONAL LEGENDRE SEED

Let (a, b, c) be positive pairwise coprime integers with $a + b = c$, and set

$$\lambda = \frac{a}{c} \in \mathbb{Q}.$$

Since $a > 0$ and $c > 0$, one has $\lambda \neq 0$. Since $b > 0$ and $a + b = c$,

$$a < c,$$

so $\lambda \neq 1$.

Define the Legendre Weierstrass equation

$$E_\lambda: y^2 = x(x-1)(x-\lambda).$$

In Weierstrass coefficients this is

$$a_1 = a_3 = a_6 = 0, \quad a_2 = -(1+\lambda), \quad a_4 = \lambda.$$

Theorem 1.1 (Discriminant).

$$\Delta(E_\lambda) = 16\lambda^2(1-\lambda)^2.$$

In particular, $\Delta(E_\lambda) \neq 0$.

Proof. The formula follows by substitution in the standard Weierstrass invariants. Every factor is nonzero because the base field has characteristic zero and $\lambda \notin \{0, 1\}$. $\square$

The Lean module `ABCFIELDCurveConstruction.lean` formalizes $\lambda = a/c$, $a < c$, $\lambda \neq 0$, $\lambda \neq 1$, the concrete curve, and nonzero discriminant. This discharges the elementary number-field/curve seed of stage one.

2. TEN-COMPONENT INITIAL THETA-DATA

The public `Iut.InitialThetaData` record groups several dependencies. The new formalization separates them as follows.

- (1) number field, algebraic closure, elliptic curve, bad locus and global conditions;
- (2) admissible prime;
- (3) actual mod- ℓ Galois representation contained in the admissible-prime record;
- (4) torsion field and its number-field/algebra tower;
- (5) orbicurves and finite covering data;
- (6) tempered fundamental-group data;
- (7) core relation;
- (8) distinguished cusp;
- (9) bad-place Tate parameters, contained in the admissible-prime record;
- (10) all remaining local theta conditions.

The dependent Lean package is

```
structure TenComponentInitialThetaPackage (AG) (TG) where
  globalCurve : GlobalFieldCurveComponent
  prime : PrimeRepresentationTorsionTateComponent globalCurve
  torsion : TorsionFieldComponent globalCurve prime
  orbicurveCoreCusp : OrbicurveCoreCuspComponent AG globalCurve prime
  temperedLocal : TemperedLocalConditionComponent
  AG TG globalCurve prime orbicurveCoreCusp
```

Theorem 2.1 (Assembly). *Every ten-component package determines an element of the exact public type*

$$Iut.InitialThetaData \text{ AG TG}.$$

Proof. Populate the public record with the dependent fields. All typeclass instances for the base field, algebraic closure, elliptic curve, torsion field and scalar tower are taken from their corresponding components. $\square$

This is an assembly theorem, not an existence theorem for the five dependent component groups.

3. AVAILABLE PUBLIC CONSTRUCTORS

Two nontrivial pieces are already available in the imported public libraries.

3.1. Tate parameters. For a complete ultrametric field K , if $\|12\| = 1$ and $1 < \|j\|$, the public Tate-curve development proves existence and uniqueness of a split multiplicative Tate parameter with invariant j . The module `ExistingFoundationConstructors.lean` uses `Classical.choose` only after this existence theorem has been proved and verifies the chosen parameter's j -invariant.

3.2. Torsion fields. Once an actual public `AdmissiblePrimeData` object is supplied, its torsion field is already proved finite over the base number field and therefore carries a public number-field instance. The remaining scalar-tower instances are made explicit in the new `TorsionFieldComponent`.

4. THE EXACT REMAINING STAGE-ONE EXISTENCE THEOREMS

For an input-indexed global curve family G_x , define

$$\begin{aligned} \text{PrimeExists}(G) &: \iff \forall x, \text{Nonempty}(\text{AdmissiblePrimeData}(G_x)), \\ \text{OrbExists}(G, P) &: \iff \forall x, \text{Nonempty}(\text{OrbicurveData}(G_x, P_x)), \\ \text{LocalExists}(G, P, O) &: \iff \forall x, \text{Nonempty}(\text{LocalThetaData}(G_x, P_x, O_x)). \end{aligned}$$

Only after these statements are proved may classical choice be used to select the pointwise family. The new Lean module `StageOneExistenceReductions.lean` formalizes this distinction.

The elementary prime-avoidance theorem is already formalized, but it does not imply the large-image Galois statement required by `AdmissiblePrimeData`. Likewise, defining the orbicurve and tempered interfaces does not prove they are inhabited by the actual geometry.

5. IUT III SOURCE INGREDIENTS

The numerical target of stage two is the public `GeneratedNativeSource`. The new source-ingredient record makes every source-facing obligation visible:

- an actual type of multiradial outputs;
- nonemptiness and one distinguished native output;
- a public generated RHS bundle;
- realization of every output as public admissible packet regions;
- an equivalence with the generated public output type;
- compatibility of the two realization maps;
- exact native q -pilot volume;
- procession-volume monotonicity on the good regions.

Theorem 5.1 (IUT III ingredient assembly). *An element of `IUTIIISourceIngredients` D Q determines an element of `GeneratedNativeSource` D Q , hence proves the public Corollary 3.12 variant for that input.*

Proof. Transport the distinguished source output through the output equivalence. Function extensionality and realization compatibility identify its public region with the source region, so the native volume equation transfers. The generated-source theorem then proves the public numerical variant. $\square$

This theorem does not create Hodge theaters, Frobenioids, Kummer/log links or the Ind1–Ind3 relation; it specifies exactly what their formalization must output.

6. IUT IV GEOMETRIC INGREDIENTS

The previous circular downstream field has been replaced by six explicit components:

- (1) encoding of an abc point into the pointwise IUT family;
- (2) actual $\log Q$, $\log \text{Diff}$ and $\log \text{Cond}$ functions;
- (3) uniform Theorem 1.10 estimate, including exceptional-set and prime-selection uniformity;
- (4) bounded q -height/canonical-height comparison;
- (5) actual different comparison;
- (6) actual conductor comparison.

The central estimate has the quantifier order

$$\forall \varepsilon > 0 \exists C_\varepsilon \forall P, \quad \text{Cor312}(P) \implies \frac{1}{6} \log Q(P) \leq (1 + \varepsilon)(\log \text{Diff}(P) + \log \text{Cond}(P)) + C_\varepsilon.$$

The constant is uniform in P .

Theorem 6.1 (IUT IV ingredient assembly). *The six ingredients determine a non-circular IUT IV bridge and imply the logarithmic abc conjecture.*

Proof. Populate the already formalized `NonCircularIUTIVBridge`. Its theorem combines the uniform q -estimate with the three comparison bounds and absorbs the fixed errors into C_ε . $\square$

7. THE REFINED FINAL THEOREM

The final dependent record now reads schematically:

```
structure RefinedFourStageProgram where
  initialTheta : Input -> Iut.InitialThetaData AG TG
  qPilot : forall x, Iut.QPilotData (initialTheta x)
  iutIII : forall x, IUTIIISourceIngredients
    (initialTheta x) (qPilot x)
  downstream : IUTIVGeometricIngredients pointwiseFamily
```

Theorem 7.1 (Refined closure).

$$\text{RefinedFourStageProgram} \implies \text{ABCConjecture}.$$

Proof. Assemble each IUT III ingredient package into the pointwise generated source family, then invoke the assembled IUT IV bridge. $\square$

The exact remaining proposition is

$$\text{Nonempty}(\text{RefinedFourStageProgram}).$$

It is not proved in the accompanying package.

8. PROGRESS TABLE

Component	Mathematical status	Lean status
$\mathbb{Q}$ and Legendre equation from an abc point	constructed	source proof
$\lambda \neq 0, 1$ and nonzero discriminant	proved	source proof
Ten-component dependent types	defined exactly	source
Assembly into public initial theta-data	proved	source proof
Tate parameter under split-multiplicative hypotheses	public theorem	wrapper proof
Torsion field after admissible prime	public theorem	wrapper proof
Large-image/admissible-prime family	open	no inhabitant

Orbicurve/core/cusp family	open	no inhabitant
Tempered/local-condition family	open	no inhabitant
IUT III source-ingredient assembly	proved	source proof
Actual Hodge-theater/Frobenioid realization	open	no inhabitant
IUT IV six-ingredient assembly	proved	source proof
Actual uniform Theorem 1.10 and comparisons	open	no inhabitant
Refined programme implies abc	proved	source proof
Refined programme inhabited	open	absent
Unparameterized <code>abc_conjecture</code>	open	absent

9. CONCLUSION

This round constructs a genuine first layer rather than another wrapper: the rational Legendre seed is explicit, the ten initial-theta components are separated and assembled, and the IUT III and IUT IV stages are decomposed into their exact non-circular ingredients. The remaining work is now concentrated in four deep existence theorems: large Galois image, actual anabelian/orbicurve/tempered geometry, the actual IUT III multiradial realization, and the uniform IUT IV geometric estimates. Until these are proved, no parameter-free proof of abc is available.

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